



Module Presenter's Manual

for

Distributed Version Control

***Effective from: March 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Module Session Allocation	1
3.	Module Deliverables on OnlineVarsity	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	7

1. Introduction

At the end of this course, students will be able to:

- Outline necessity for Version Control.
- Outline an Overview of Git.
- Describe basics of Git.
- List Git terminology.
- Describe how to make a new Git repository.
- Describe how to make a clone of an existing Git repository.
- Define configuration of Git.
- Outline staging and committing changes.
- Identify how to create a new branch.
- Outline steps to develop code on multiple branches.
- Outline steps to modify a branch.
- Identify how to merge and delete branches.
- Describe how to create a remote branch.
- Describe how to pull from remote connection.
- Explain GitHub.
- Differentiate between Git and GitHub.
- Describe how to create a GitHub repository.
- Describe Branching, Committing, and Pulling in GitHub.
- Describe forking and other operations with GitHub.
- Identify the process to clone a repository.
- Explain GitHub Pages.
- Describe different types of GitHub Pages Sites.
- List the guidelines for using GitHub Pages.

2. Information on Session Allocation

Module	Online (No. of Hrs)	Self-study (No. of Hrs)
Distributed Version Control	12	8

Throughout this Presenter's Manual, the module **Distributed Version Control** will be referred to as **Git & GitHub**.

3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2		Day 3	Day 4	
1	Session 1 Git & GitHub – TL1	Session 2 Git & GitHub – TL2		Session 3 Git & GitHub – TL3	Session 4 Git & GitHub – TL4	S1-S2
2	Session 5 Git & GitHub – TL5	Session 6 Git & GitHub – TL6	S3 -S4			

TL: Online Session

S: Self-study Hours (over and above the module duration)

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	Git & GitHub– TL1	All the concept lessons as listed below from Session 1 of <i>Version Control with Git and GitHub</i> book should be covered in this session. <u>Session 1 – Getting Started with Git</u> ➤ Necessity for Version Control ➤ Introduction to Git ➤ Git Basics ➤ Command-Line Program ➤ Setting Up Git	<u>Version Control with Git and GitHub</u> SG - Session 1 XP - Session 1 TG - Session 1
2	Git & GitHub– TL2	All the concept lessons as listed below from Session 2 of <i>Version Control with Git and GitHub</i> book should be covered in this session. <u>Session 2 – Working with Git</u> ➤ Getting a Git Repository ➤ Setting Up the Project Folder ➤ Git Configuration ➤ Levels of Configuration ➤ Git Clone ➤ Staging Files ➤ Commit Files	<u>Version Control with Git and GitHub</u> SG - Session 2 XP - Session 2 TG - Session 2

Session No.	Session Title	Session Details	Deliverables' Mapping
3	Git & GitHub – TL3	All the concept lessons as listed below from Session 3 and Session 4 of <i>Version Control with Git and GitHub</i> book should be covered in this session. <u>Session 3 – Git Branching</u> ➤ Basic Branching and Merging ➤ Creating a New Branch ➤ Delete, Rename, and Merge two Branches ➤ Basic Merging to Master Branch <u>Session 4 – Different Approaches to Git</u> ➤ Remote Branches in Git ➤ Pulling in Git	<u>Version Control with Git and GitHub</u> SG - Session 3 & 4 XP - Session 3 & 4 TG - Session 3 & 4
4	Git & GitHub – TL4	The workshop lessons of Session 3 and Session 4 of <i>Version Control with Git and GitHub</i> course should be covered in this session.	<u>Version Control with Git and GitHub</u> <u>(OnlineVarsity)</u> Session 3 & 4
	S1-S2	After the session, ask the students to refer the References and solve Assignments of Session 3 and Session 4 available on Onlinevarsity.	<u>Version Control with Git and GitHub</u> <u>(OnlineVarsity)</u> Session 3 & 4

Session No.	Session Title	Session Details	Deliverables' Mapping
5	Git & GitHub – TL5	All the topics as listed below from Session 5 and Session 6 of <i>Version Control with Git and GitHub</i> book should be covered in this session. <u>Session 5 – GitHub as a Project Repository</u> ➤ Introduction to GitHub ➤ GitHub Web Interface ➤ Differences between Git and GitHub ➤ Installing GitHub Desktop ➤ Creating a Repository in GitHub ➤ Branching in GitHub ➤ Viewing Branches ➤ Deleting Branches ➤ Other Operations ➤ Committing in GitHub <u>Session 6 – More on GitHub and Overview of GitHub Pages</u> ➤ GitHub Pricing Plans ➤ Forking and Other Operations in GitHub ➤ Cloning a Repository ➤ Introduction to GitHub Pages ➤ Types of GitHub Pages Sites ➤ Guidelines for using GitHub Pages	<u>Version Control with Git and GitHub</u> SG - Session 5 & 6 XP - Session 5 & 6 TG - Session 5 & 6
6	Git & GitHub – TL6	The workshop lessons of Session 3 and Session 4 of <i>Version Control with Git and GitHub</i> course should be covered in this session.	<u>Version Control with Git and GitHub</u> <u>(OnlineVarsity)</u> Session 3 & 4
	S3-S4	After the session, ask the students to refer the References and solve Assignments of Session 5 & Session 6 available on Onlinevarsity.	<u>Version Control with Git and GitHub</u> <u>(OnlineVarsity)</u> Session 5 & 6

6. Library References

➤	Git by Ryan Hodson
➤	The Ultimate Beginners Guide Git by Adney Ainley
➤	Building Tools with GitHub by Rosemie Callewaert and Johan Revilla
➤	Version Control with Git by Jon Loeliger, Matthew McCullough

~~ End of Document ~~